Experiment 1:

Optical Measurements

Authors: Eli Raz, Itamar Hason

Motivation: Rainbow

Rainbow phenomenon, result of:

- Minimal deflection angle off a water drop
- Refractive index dependence on the wavelength

Use the **minimal deflection angle** phenomenon to obtain the optical parameters of a disk, a prism and a diffraction grating with **superb precision**



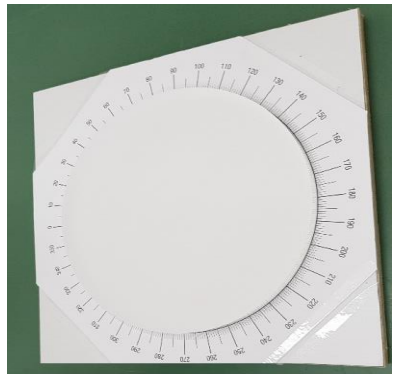
The Leading Principle: Extremum Experimental Method

- Ordinary experimental method:
  - Measure experimental variables
  - Extract physical quantity: solving equations, regression
  - Sensitive to errors in **all** experimental variables
- Alternative method: measure at an extremal setting
  - Achievable with great precision
  - Supplies further equations: solve for less-easy-to-measure variables
  - Extract physical quantity using **subset** of the experimental variables

Parts of the Experiment

Part A

- Disk, Mimicking the rainbow effect, with one wavelength
- Finding refractive index with **superb precision**
- Multiple orders



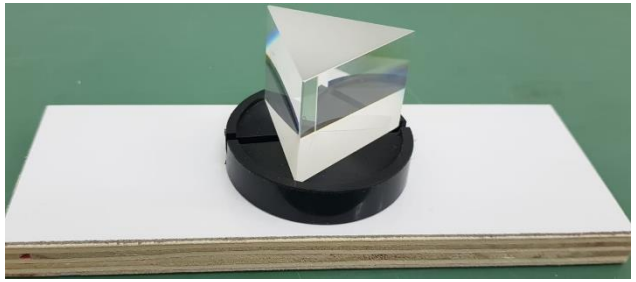
Part B

- Diffraction grating
- Finding the ratio λ/d with **superb precision**
- Multiple orders



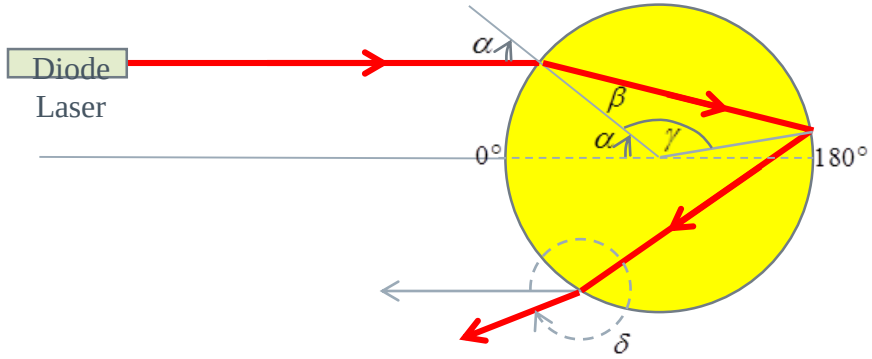
Part C

- Prism
- Finding refractive index with **superb precision**



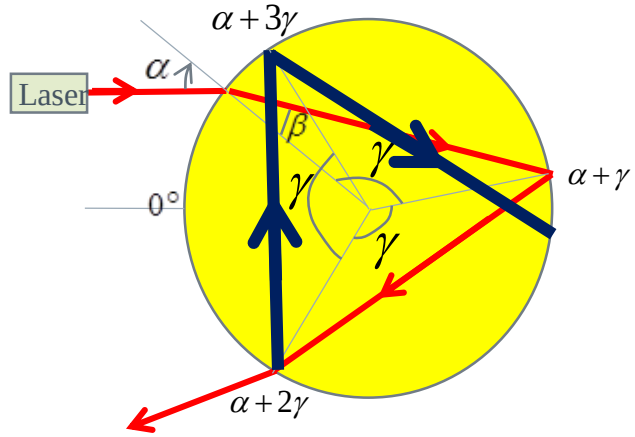
Part A: The refractive index of a disk

- Measure deflection angle as a function of the incident angle
- **Snell's law** with regression is possible, suffers from **bad precision** for the incident angle due to the **width of the ray**
- **Minimal deflection angle** is easy to find despite the width of the ray
- Refraction angle only is enough at the minimal deflection angle



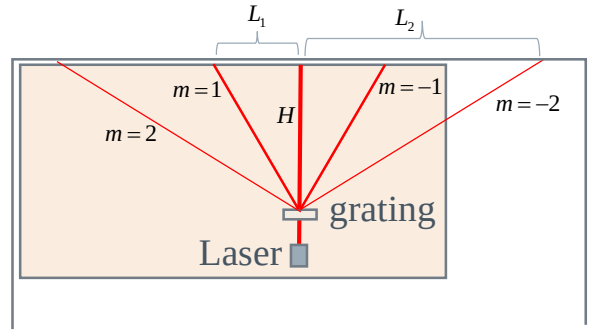
Part A: The refractive index of a disk

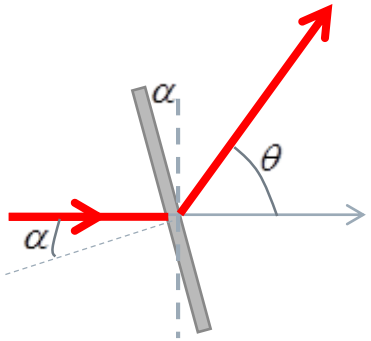
- Use **multiple reflections** to find the refraction angle with high precision
- Obtain the refractive index with super precision
- Analysis is possible with "**higher order rainbows**" as well
- Perform the same analysis on different rainbow orders to get an estimate on the error of the refractive index (<0.5%)

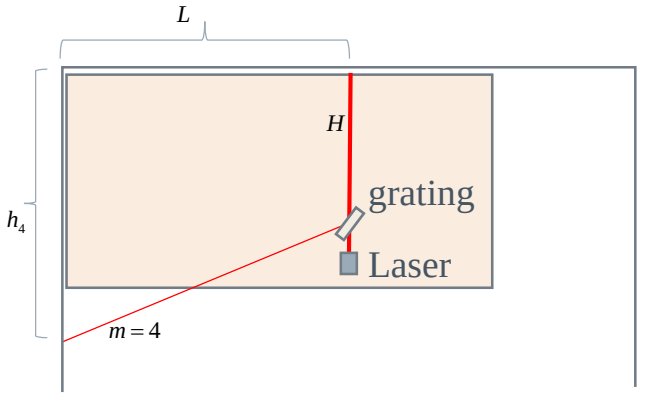


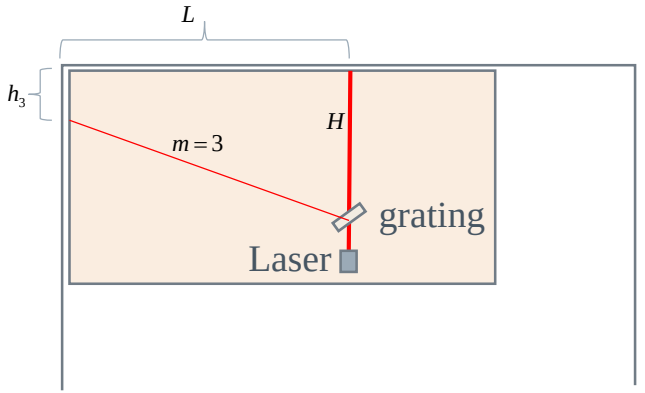
Part B: Diffraction grating parameters

- Regular method requires knowing the incident angle α which is difficult to measure with good precision
- Finding the incident angle where the maximum of order m is at a minimal deflection angle, gives a second equation which allows to avoid measuring the incident angle
- The students are required to repeat the measurements at various maxima orders



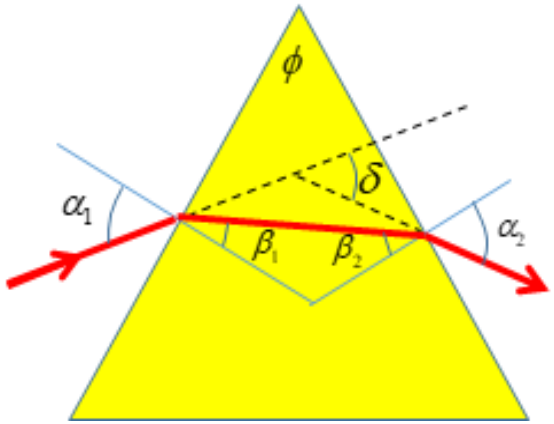


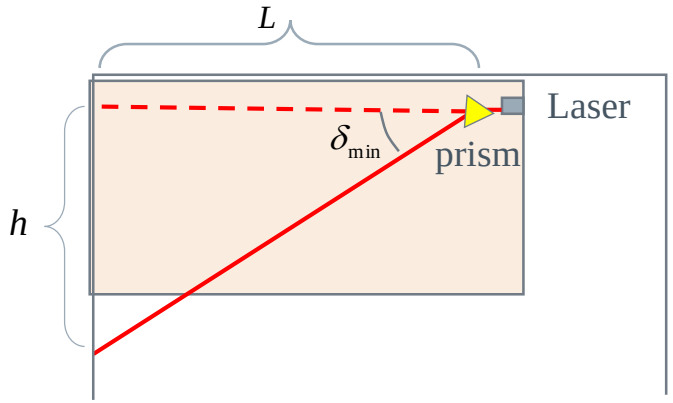




Part C: The refractive index of a prism

- Again, measuring the incident angle is difficult
- Using the minimal deflection angle allows to relate the incident angle and the deflection angle, overcoming the difficulty in measuring the incident angle





Summary

- **Original** and **sophisticated** experimental method
- **Superb precision** with **minimal obstacles**
- Direct methods – **possible but less accurate**

- Students are tested for: **Creativity, Physical understanding, Experimental technique**

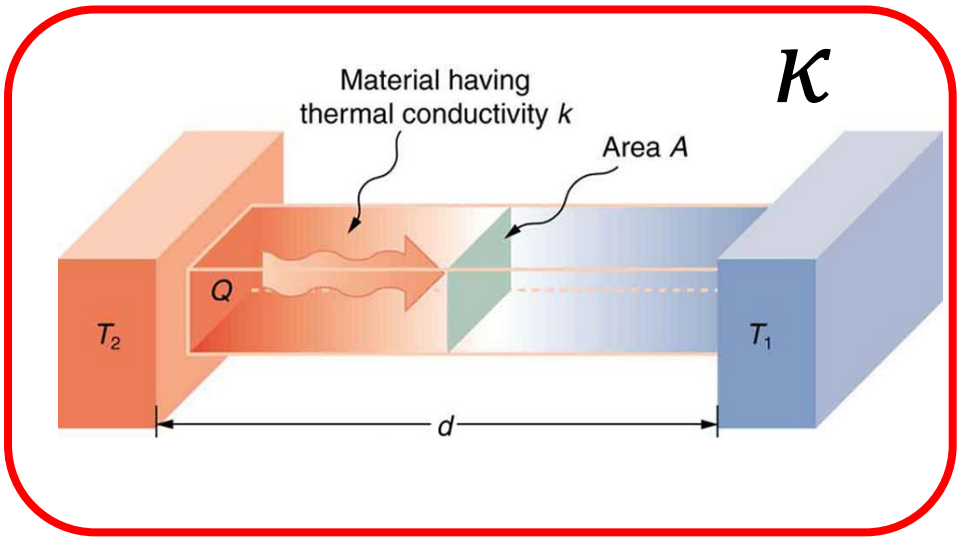
The Wiedemann-Franz Law

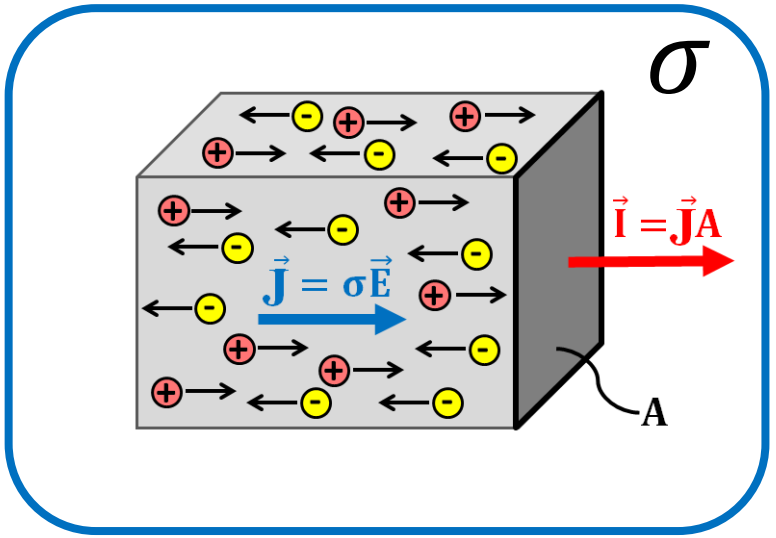
Asaf Rozen, Alexander Palevski, Eli Raz

The Wiedemann-Franz Law

- Heat and electrical transport involve free electrons

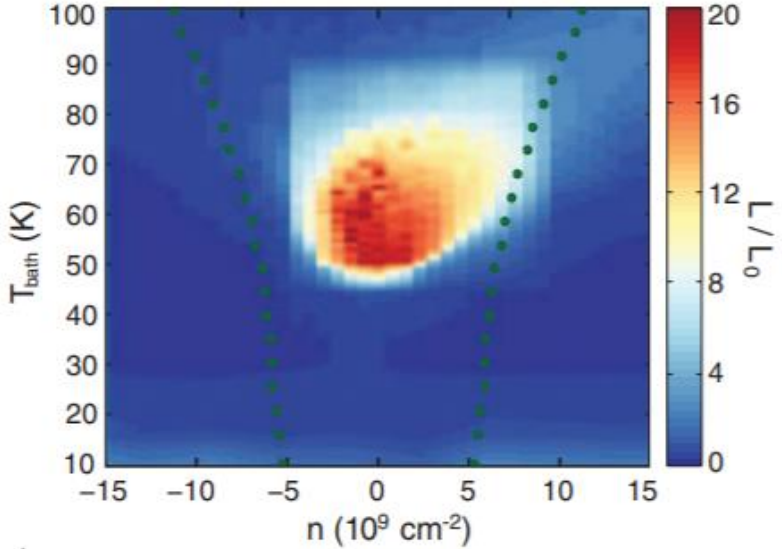
$$L = \frac{\kappa}{\sigma T} = \frac{\pi^2}{3}\left(\frac{k_B}{e}\right)^2 = 2.44 \cdot 10^{-8}\left[\frac{W\Omega}{K^2}\right]$$

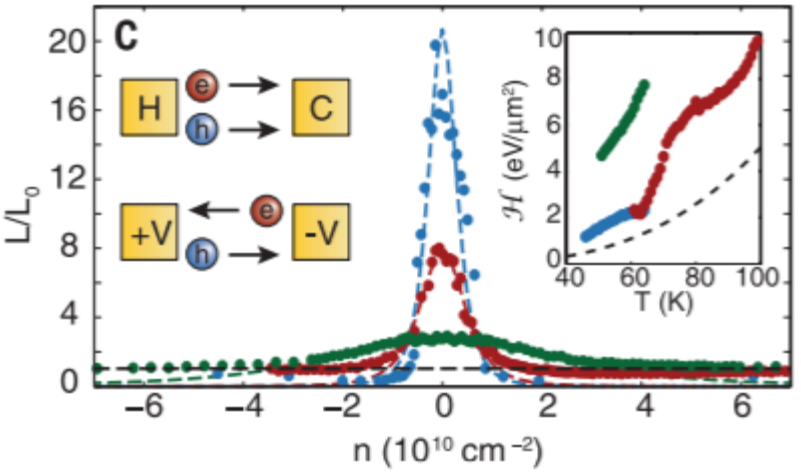




The Wiedemann-Franz Law

- Discovered in 1853 by Gustav Wiedemann and Rudolph Franz
- Broadly used in modern solid state physics

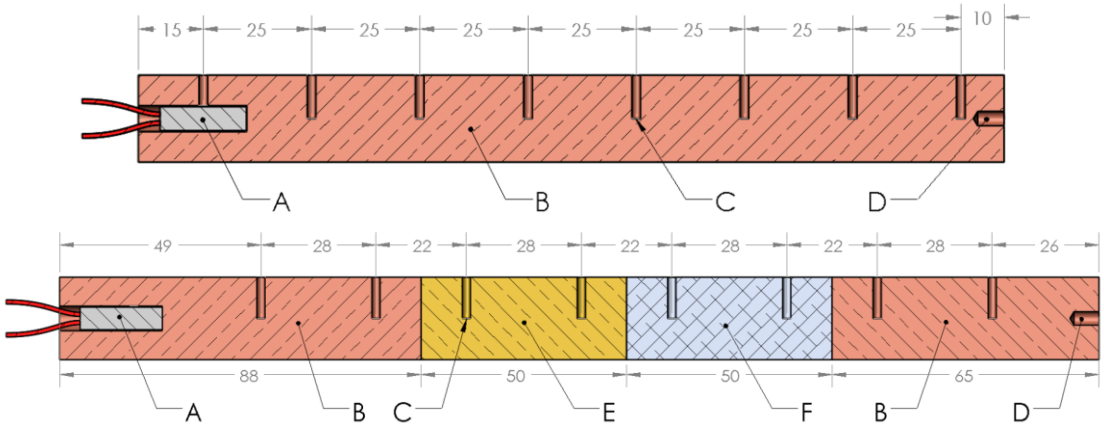




Crossno, J., Shi, J. K., Wang, K., Liu, X., Harzheim, A., Lucas, A., ... & Ohki, T. A. (2016). Observation of the Dirac fluid and **the breakdown of the Wiedemann-Franz law** in graphene. ***Science***, *351*(6277), 1058-1061.

Parts Of The Experiment

- Part A – Electric conductivity (σ) of various metals.
- Part B – Thermal conductivity (κ) of Copper.
- Part C – Measure heat capacity (c_p) and heat loss.
- Part D – Measure the relative κ of different metals.
- Part E – Discover the Wiedemann-Franz law.



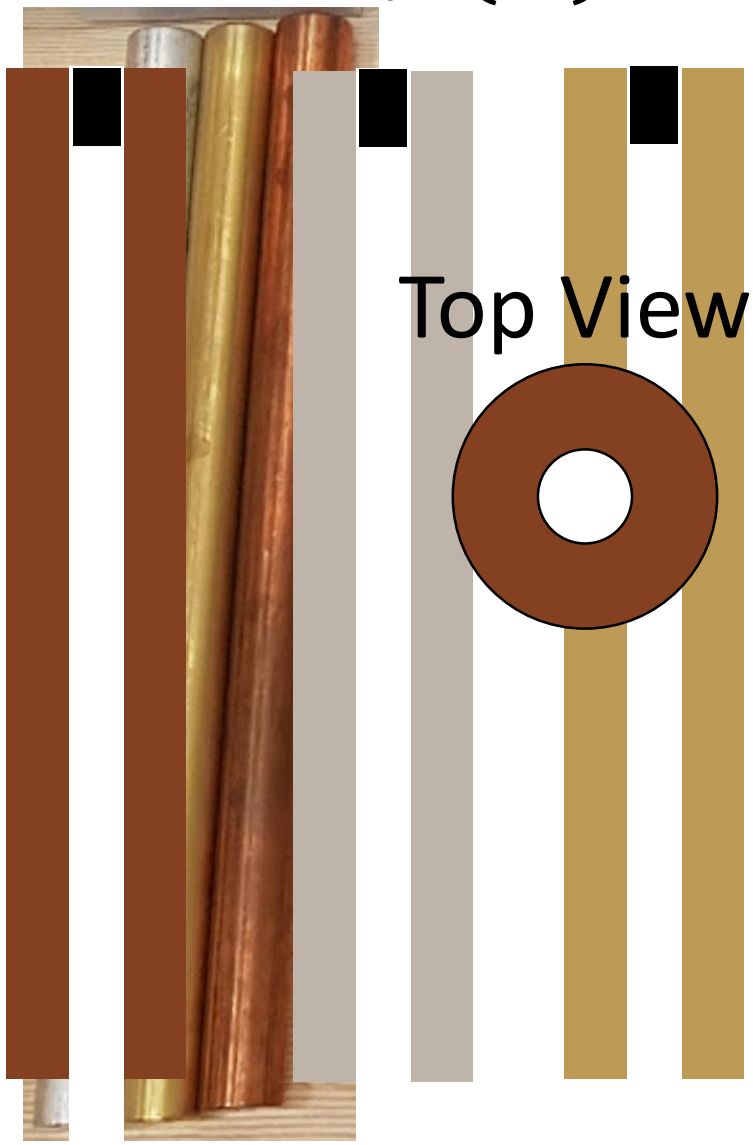
Equipment



Part A – Electrical conductivy (σ)

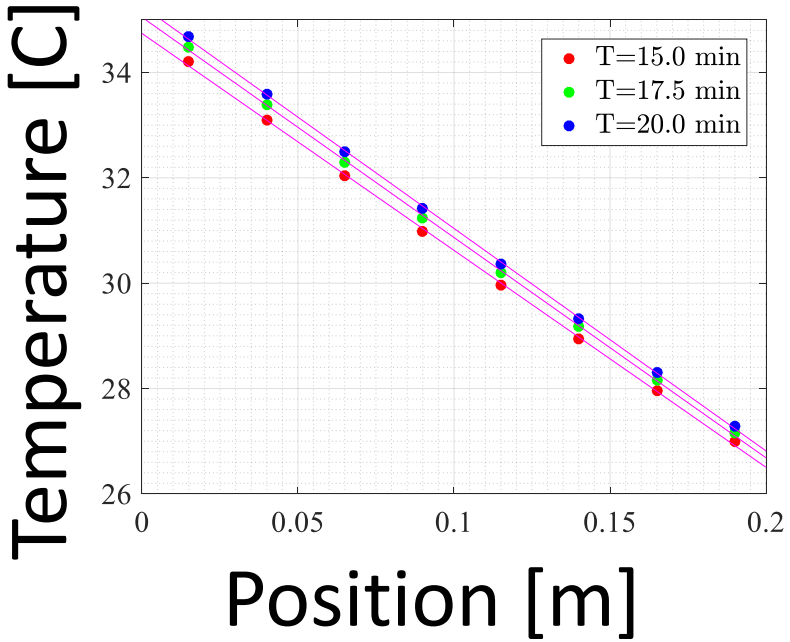
- Aluminum, Brass and Copper tubes
- Small magnet falls at a constant speed

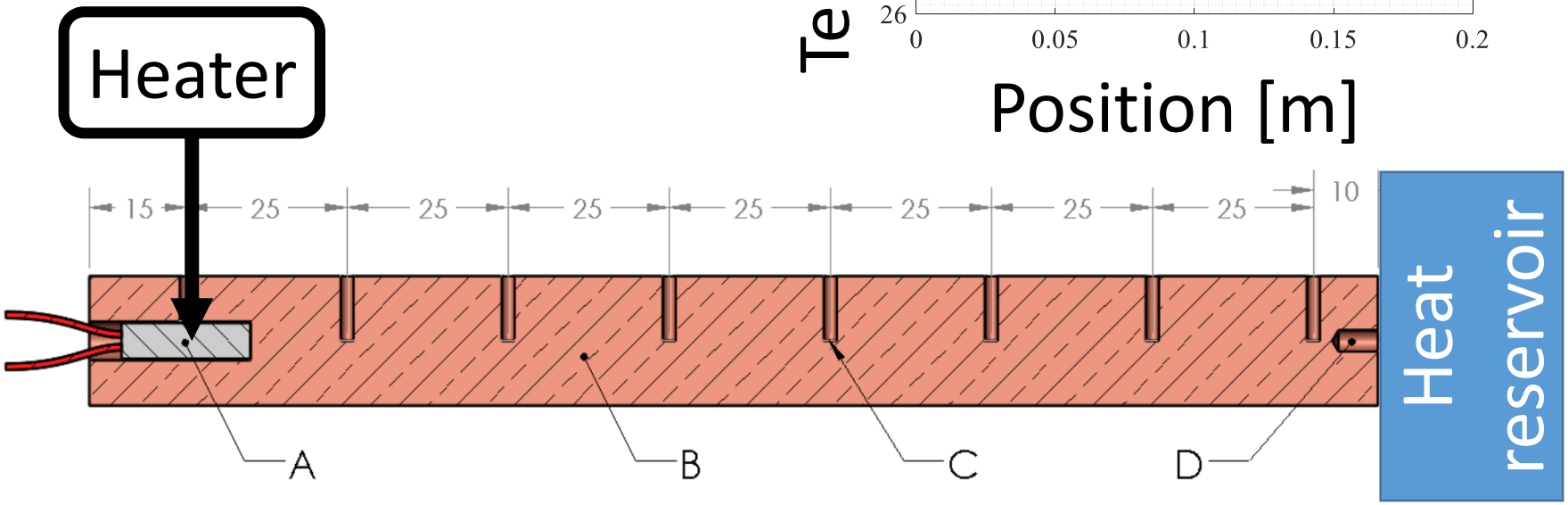
$$t = 0.22 \frac{\pi r_m^2 (\mu_0 M)^2 w L_0}{mg} \sigma$$



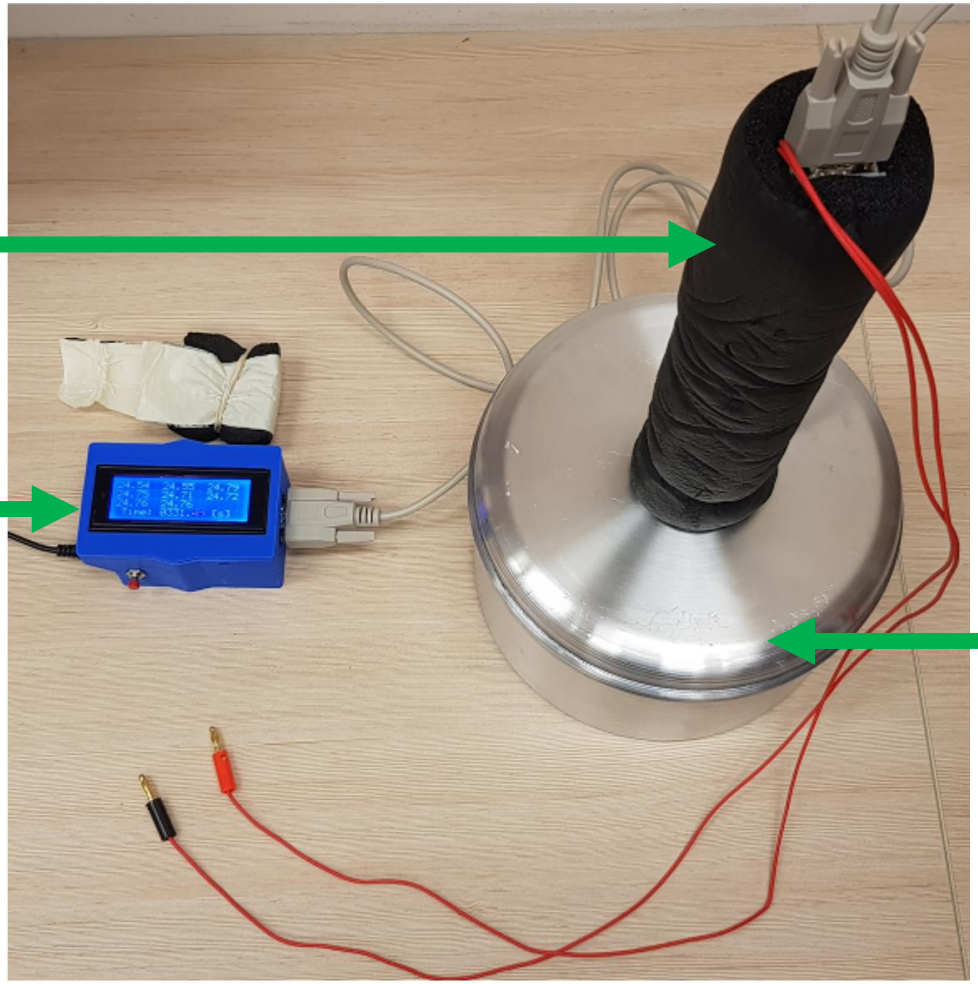
Part B – Thermal conductivity (κ)

- Power on one side, Reservoir on the other





Part B – Thermal conductivity (κ)

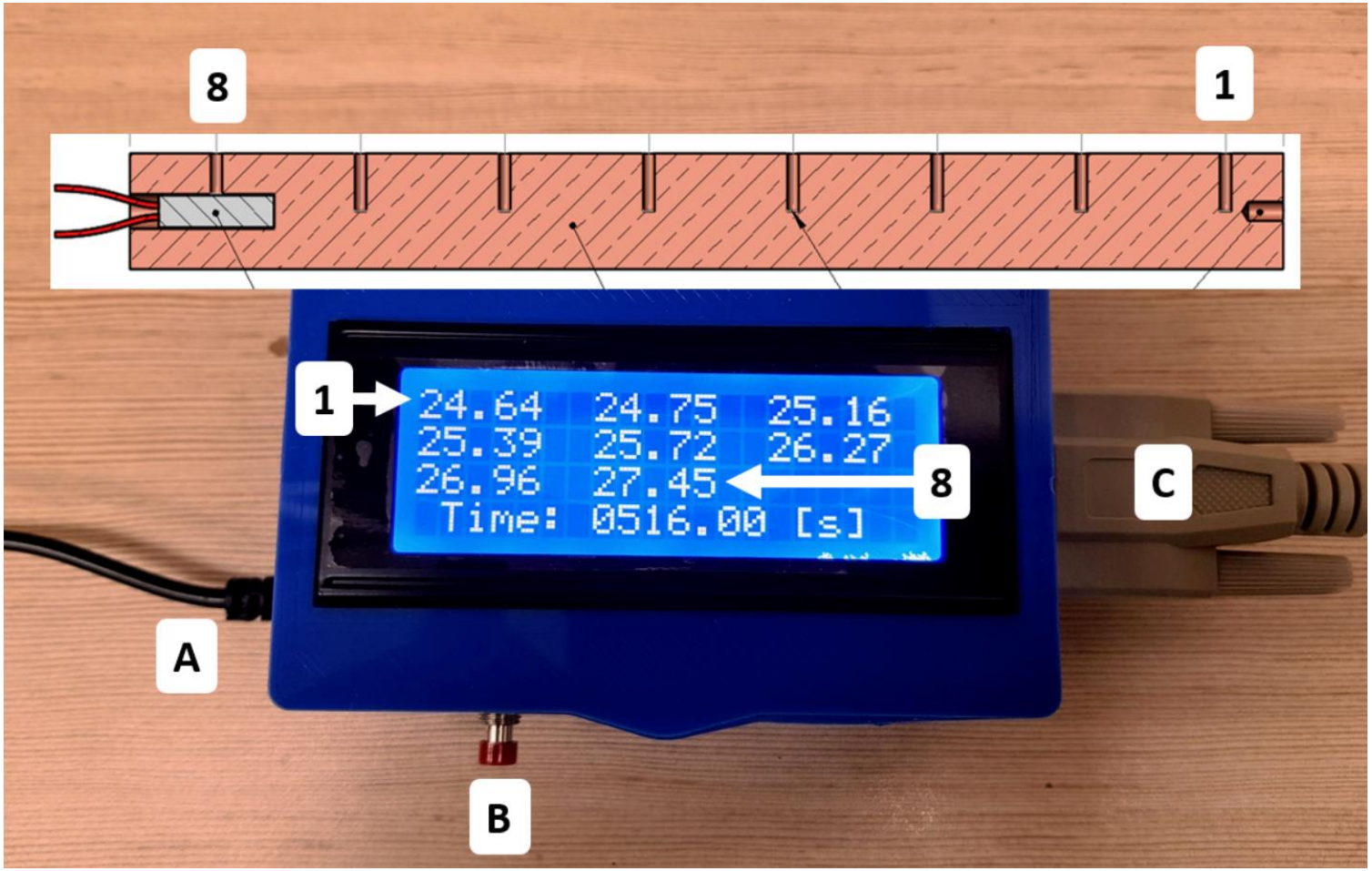


Isolated Rod

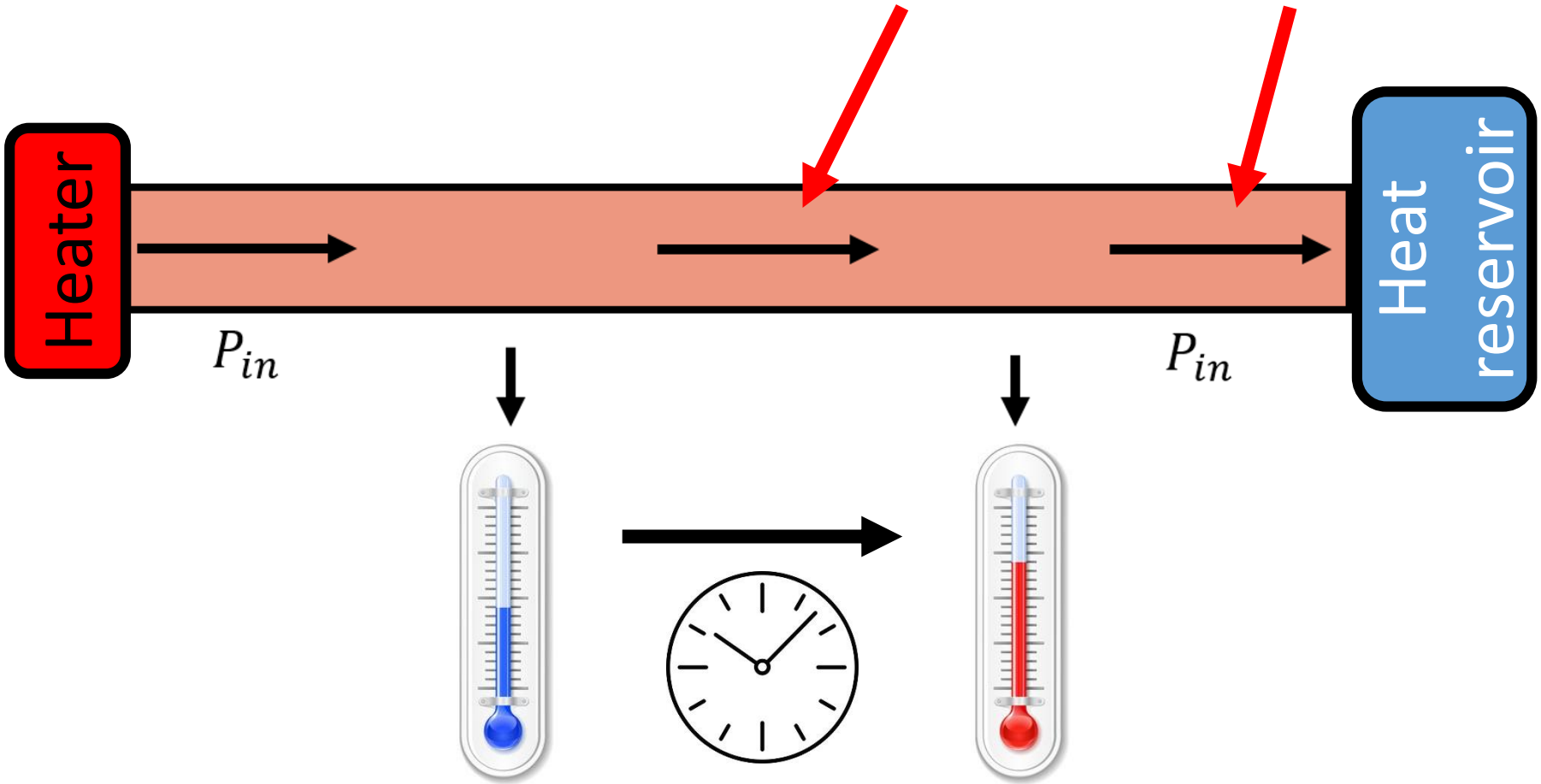
Readout Box

Heat Reservoir

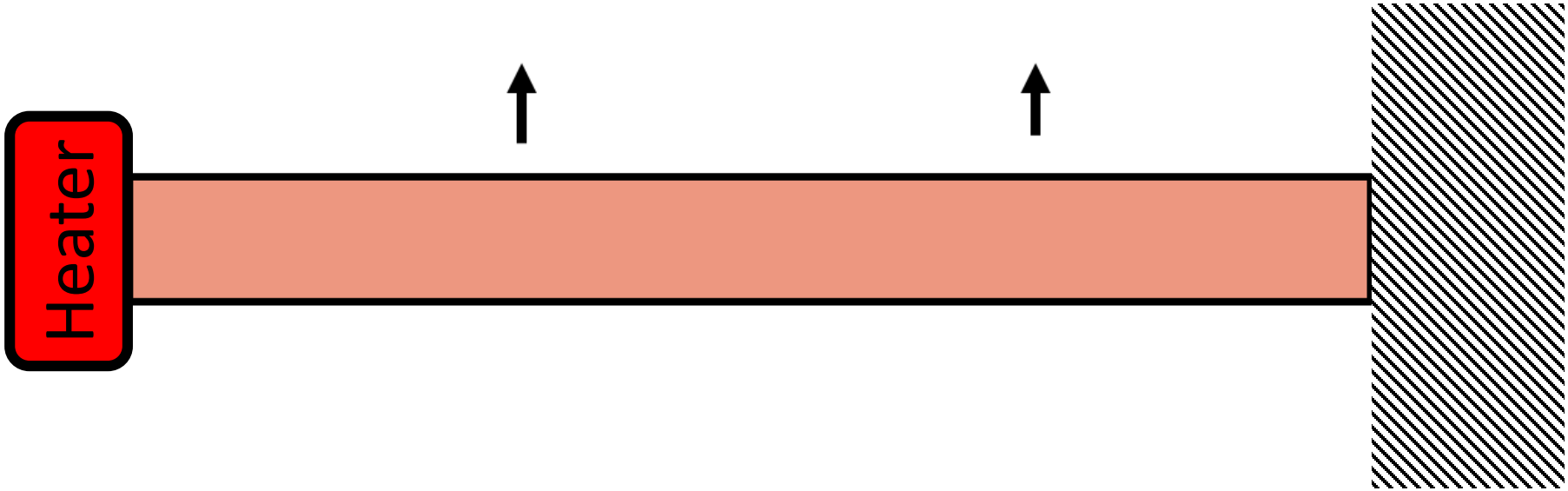
Part B – Thermal conductivity (κ)



Part C – Measure Heat Capacity And Heat Loss To Correct κ

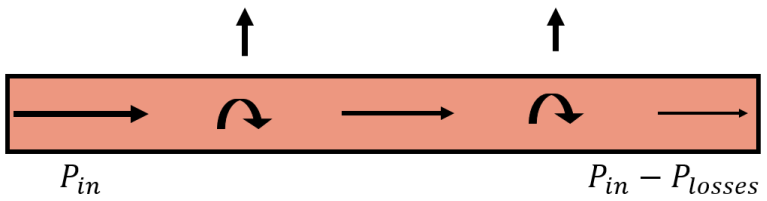


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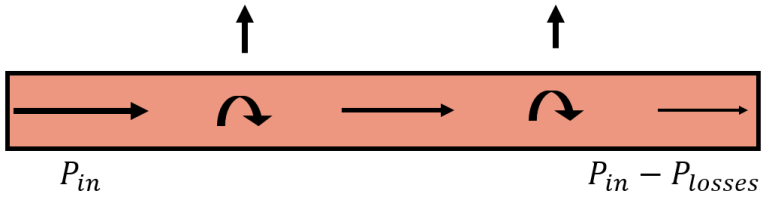
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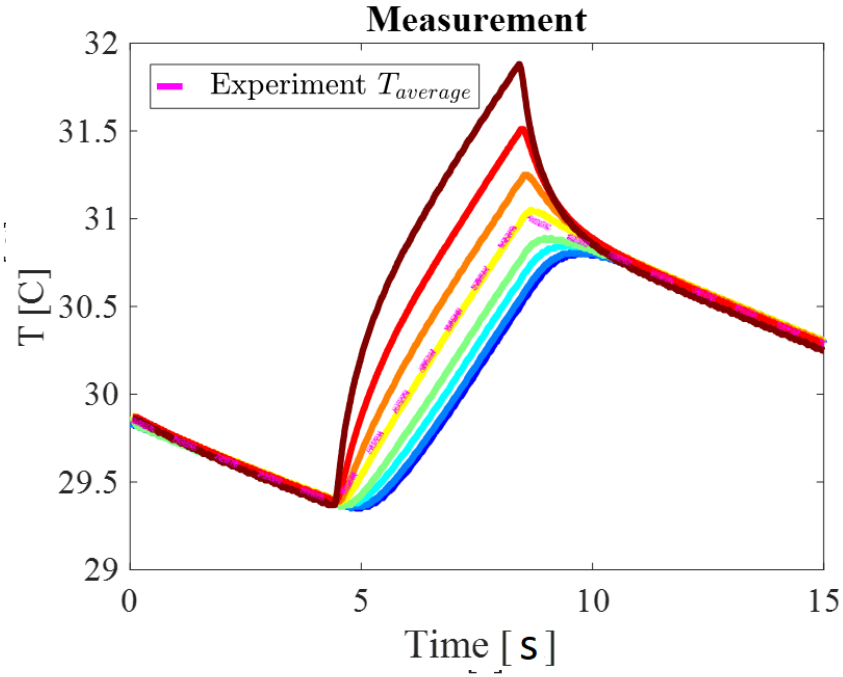
- The rod is not at steady state
- There is heat loss to the environment



Part C – Measure Heat Capacity And Heat Loss To Correct κ

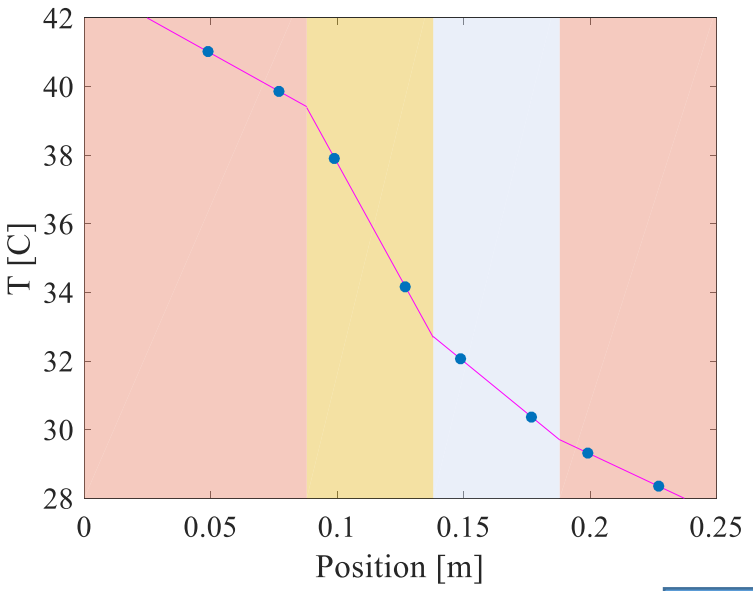
- The rod is not at steady state
- There is heat loss to the environment
- We use cooling-heating-cooling cycle

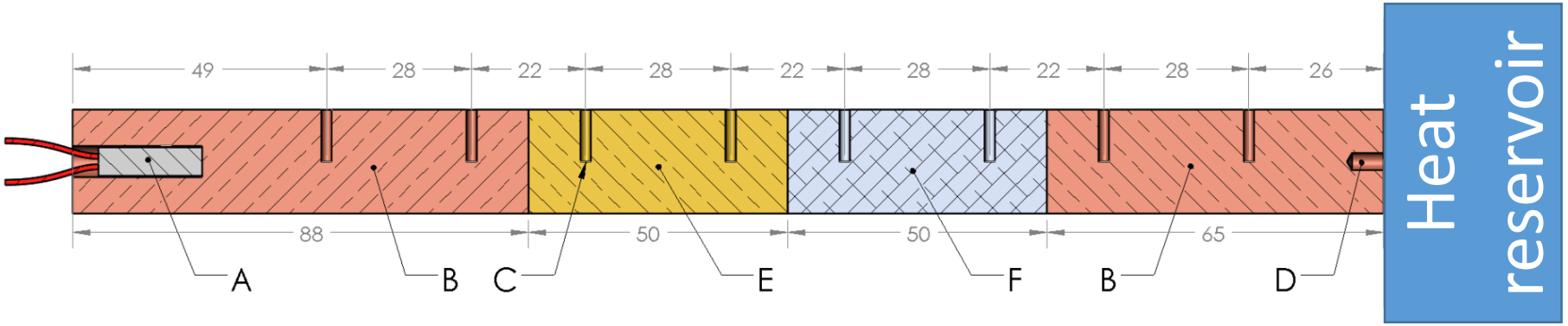




Part D – Measure The Relative κ Of Different Metals

- Using alternating material rod
- Get accurate relative thermal conductivity
- Using the knowledge from part C to correct results





Part E – Discover The Wiedemann-Franz Law

Material / Property	Copper	Aluminum	Brass
$\sigma\ [\Omega^{-1}m^{-1}]$ Electrical conductivity	5.97×10^{7}	2.89×10^{7}	1.60×10^{7}
$\kappa \left[\frac{W}{Km}\right]$ Heat conductivity	396	239	115
$L \left[\frac{W\Omega}{K^2}\right]$ Lorenz coefficient	2.23×10^{-8}	2.27×10^{-8}	2.42×10^{-8}

Acknowledgments – Roy Beck- Barkai, Yoram Dagan